Modelling combined UV/chlorine treatment of filtered water

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1 Introduction

Treatment of water for human consumption consists of several steps: coagulation, flocculation, sedimentation, filtration, and disinfection, as illustrated in Figure 1. In this report, we focus on disinfection, modelling how pathogen populations evolve when exposed to a mixture of water-dissolved chlorine and UV light.

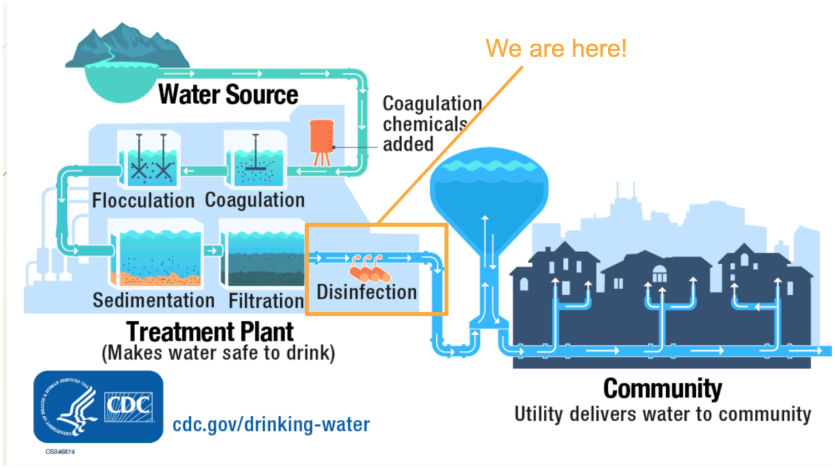


Figure 1: Typical layout of a water treatment plant.

We show that the model we developed combined with scaling constants taken from literature is representative of a real-world system. Therefore, our model can be used to answer a broad range of questions regarding the system parameters such as how much chlorine is needed, what the pathogen population will look like, or how specific pathogens resistant to disinfectants may behave, which can be used for academic or commercial purposes.

1.1 Previous works

There is an extensive corpus of scientific literature regarding single-method disinfections (such as chlorine, mono-chloramine, UV, and ozone), but combined methods are still quite new. We used Kishimoto's 2019 overview of combined UV/chlorine disinfection processes [1] as our basis for understanding how chlorine and UV interact with pathogens. Furthermore, to test our model on a wide range of parameters, we used a database of pathogen treatment results compiled by Mofidi et al. [2] to cross-reference whether our model behaves as we would expect.

2 Disinfection mechanisms

2.1 Chlorine

Water treatment plants usually add chlorine to water in one of two ways. The first is to use chlorine gas Cl_2 that was liquefied under high pressure that produces hypochlorous acid and ions directly

$$Cl_2 + H_2O \rightarrow HOCl + OCl^-$$

The second method is dissolving hypochlorous salts—sodium hypochlorite NaOCl or calcium hypochlorite $Ca(OCl)_2$ with water, which primarily produces hypochlorous acid and hydroxides

$$NaOCl + H_2O \rightarrow HOCl + NaOH$$

which then further reacts with the surrounding water to produce hypochlorite ions OCl^-. For our model, we assume the use of pressurised chlorine gas, since it takes less time for it to mix and react with water and achieve an equilibrium concentration. Furthermore, it simplifies our model by not having to account for sodium and/or calcium hydroxide.

Hypochlorous compounds are highly reactive with organic matter, which makes them highly effective at neutralising pathogens. However, they also react with natural organic matter (NOM) in ways which can produce toxic or carcinogenic byproducts [1]. Therefore, ensuring that the chlorine dose is sufficient to neutralise pathogens without remaining in the supply is crucial for safety as well as taste.

2.2 Ultraviolet light

UV light neutralises pathogens by damaging their genetic material (DNA or RNA), making them unable to cause infection and reproduce. Research suggests that monochromatic light with a peak at 265 nm has the highest efficacy [1]. UV light is often supplied by monochromatic (narrow-spectrum) bulbs placed in tubes within the water tank, thereby mitigating the loss of intensity caused by the power-to-area ratio, absorption, and scattering with water.

2.3 Combining chlorine and ultraviolet light

Introducing two disinfectants into water has a synergistic effect, when compared to single-disinfectant methods. In our case, pathogens that have been irradiated are more vulnerable to chlorine since they are unable to repair themselves. What is more, ultraviolet light photolyses water-dissolved chlorine compounds into radicals: highly reactive compounds that have an unpaired electron in the valence shell. Specifically, in our model we look at first-order radicals of hypochlorous acid HOCl.

$$\mathrm{HOCl} + \gamma_{254\mathrm{nm}} \rightarrow \mathrm{HO}\bullet + \mathrm{Cl}\bullet$$

Interestingly, the highest conversion rate to chlorine radicals occurs for ultraviolet light at a wavelength of 254 nm, which is quite close to the 265 nm peak of bacterial inactivation. Since the most readily-available data is for these two values, in our model we assume that the UV lamp used is approximately monochromatic at 260 nm with a peak width of 10 nm, thereby ensuring both maximal photolysis and maximal inactivation.

3 Our model

In our model, we track a chlorine concentration C and photolysed chlorine radical concentration P with units of $\mathrm{kg/m^3}$, and a pathogen concentration B in colony-forming units per volume $\mathrm{CFU/m^3}$. In literature, pathogen concentrations are often measured in CFU-per-volume rather than counts-per-volume since pathogens like to clump together, appearing as a single unit for petri dish measurements.

For analysis and for our simulations, we rescale our model to be dimensionless

$$B = B_0 b \qquad C = C_0 c \qquad P = C_0 p \qquad t = \tau s$$

with a shared scale for chlorine and photolysed chlorine C_0, and a characteristic timescale τ set by the interaction of bacteria and chlorine.

$$\tau = \frac{1}{k_{BC} C_0} \qquad \text{since} \qquad b_s = -\underbrace{\tau k_{BC} C_0}_{=1} bc + \ldots$$

3.1 Reduced models

To ensure our models behave like expected, we started with two simplified cases.

The first was a chlorine-only model where pathogens reproduce at a rate k_B and die to chlorine at a rate k_{BC}; and chlorine concentrations depend on their interaction with the environment k_C and its reaction with pathogens.

$$\begin{aligned}\frac{\mathrm{d}B}{\mathrm{d}t} &= \quad k_B B - k_{BC} BC \\ \frac{\mathrm{d}C}{\mathrm{d}t} &= -k_C C - k_{BC} BC\end{aligned}$$

For the second model, we also included the interaction between chlorine and UV light at a rate k_{IC} and an interaction between pathogens and UV light at a rate k_{BI} (i.e. neglecting photolysis).

$$\begin{aligned}\frac{\mathrm{d}B}{\mathrm{d}t} &= \quad k_B B - k_{BC} BC - k_{BI} BI \\ \frac{\mathrm{d}C}{\mathrm{d}t} &= -k_C C - k_{BC} BC - k_{IC} CI\end{aligned}$$

3.2 Full model

For our full model, we also include the interactions of pathogens and chlorine with photolysed radicals.

$$\begin{aligned}\frac{\mathrm{d}B}{\mathrm{d}t} &= -B(-k_B + k_{BC} C + k_{BP} P + k_{BI} I) \\ \frac{\mathrm{d}C}{\mathrm{d}t} &= -C(k_C + k_{IC} I + k_{BC} B) \\ \frac{\mathrm{d}P}{\mathrm{d}t} &= k_{IC} IC - P(k_P + k_{BP} B)\end{aligned}$$

When we rescale using our characteristic timescale τ, we obtain a dimensionless model in b, c, p, where we reduce our constants into $\delta_1 ... \delta_6$:

$$\begin{aligned}\frac{\mathrm{d}b}{\mathrm{d}s} &= b\left(\underbrace{\frac{k_B}{k_{BC}C_0} - \frac{k_{BI}I}{k_{BC}C_0}}_{=\delta_1} - \underbrace{\frac{k_{BP}}{k_{BC}}}_{=\delta_2} p - c\right) \\ \frac{\mathrm{d}c}{\mathrm{d}s} &= -c\left(\underbrace{\frac{k_C}{k_{BC}C_0} + \frac{k_{IC}I}{k_{BC}}}_{=\delta_3} + B_0 b\right) \\ \frac{\mathrm{d}p}{\mathrm{d}s} &= \underbrace{\frac{k_{IC}I}{k_{BC}C_0}}_{=\delta_4} c - p\left(\underbrace{\frac{k_P}{k_{BC}C_0}}_{=\delta_5} + \underbrace{\frac{k_{BP}B_0}{k_{BC}C_0}}_{=\delta_6} b\right)\end{aligned}$$

Here, we arrive at an issue: the model is not truly dimensionless, since the units of B and C are not the same, but we have the same term $k_{BC}BC$ in both of these equations. To resolve this issue, we looked at experimental data, and found that the reduction of chlorine due to

bacteria is $\mathcal{O}(10^5)$, whereas our model was rescaled to be $\mathcal{O}(1\text{–}10)$ (more details in Table 1). Therefore, we neglect the $k_{BC}BC$ term in the equation for C, and find that the expression becomes analytically solvable as exponential decay with some constant δ_3.

$$c(s) = c(0)e^{-\delta_3 s} = \exp(-\delta_3 s) \qquad \text{where } c(0) = 1$$

We then substitute this into our equations for b and p to obtain our final, two-equation system:

$$\frac{\mathrm{d}b}{\mathrm{d}s} = -b\left(-\delta_1 + \delta_2 p + \underbrace{\exp(-\delta_3 s)}_{=c}\right)$$

$$\frac{\mathrm{d}p}{\mathrm{d}s} = \delta_4 \underbrace{\exp(-\delta_3 s)}_{=c} - p(\delta_5 + \delta_6 b)$$

3.3 Assumptions and limitations

PATHOGENS

For this model, we assume B is a mix of bacteria and viruses. We also neglect bacteriophages, i.e. our viruses do not reproduce, and k_B is strictly set by the reproduction rate of bacteria. Moreover, while our model can also model protozoa, our database [2] only had data on viruses and bacteria.

FILTERED WATER

As mentioned in Section 2.1, chlorine can interact with natural organic matter to create toxic byproducts. Therefore, we assume our water is perfectly filtered and only includes pathogens. In practice, since we reduced chlorine to an analytical function, we can check for the concentrations of toxic chlorine byproducts by looking at the chlorine concentration directly. However, since we followed World Health Organisation guidelines [3] for setting our chlorine values (see Table 1), we decided to neglect the byproducts.

HOMOGENEITY AND ISOTROPY

Since most of our data comes from lab experiments consisting of a petri dish with almost-perfect irradiation, we assume that our system is perfectly homogeneous and isotropic. We discuss this further in Section 5.1.

DISSOLVED CHLORINE

We assume that at $t = s = 0$ (when the ultraviolet light is turned on), chlorine is already fully dissolved and perfectly dispersed. This means that our starting pathogen population is smaller than in practice, since some chlorine and pathogens have already reacted. However, due to the timescales of chlorine decay being in tens of hours and dissolving happening in seconds, we consider this effect to be negligible.

TEMPERATURE AND ACIDITY

We assume a standard pH of 7.5 and a temperature of $T = 20\,^{\circ}\mathrm{C}$, which are standard parameters for drinking water. We further assume that these parameters do not change since the concentrations of chlorine are very low (see Table 1), therefore we can neglect the acidic and exothermic byproducts of dissolving chlorine.

SYSTEM GEOMETRY

We assume a reactor-based system, as opposed to a flow-through tube system, which allows us to set the maximum time $t_{\max}$ as a free parameter. In practice, if a tube is long enough

and narrow enough to properly expose pathogens to disinfectants, flow-through systems can be more efficient at disinfection (see Section 4 and Section 5.1 for more discussion on this topic).

3.4 Constants

symbol	description	value	units	source
B_0	initial bacterial concentration	1.6×10^5	CFU m^{-3}	[4]
C_0	initial dissolved chlorine concentration	2.5×10^{-3}	kg m^{-3}	[3,5]
$C(t_{\max})$	acceptable final chlorine concentrations	2–5 $\times 10^{-4}$	kg m^{-3}	[3]
$t_{\max}$	typical length of treatment	10^3–10^4	s	[6]
k_B	bacterial growth rate	5.5×10^{-6}	s^{-1}	[7]
k_C	chlorine (HClO + ClO^-) decay	4.3×10^{-5}	s^{-1}	[8]
k_P	photolysed radical ($HO\bullet, Cl\bullet$) decay	1.4×10^5	s^{-1}	[9]
k_{IC}	production of photolysed chlorine via UV	2.9×10^{-5}	$m^2\ W^{-1}\ s^{-1}$	[10,11]
k_{BI}	bacterial inactivation via UV	5.0×10^{-2}	$m^2\ W^{-1}\ s^{-1}$	[2,12]
k_{BC}	bacterial inactivation via chlorine	8.5	$m^3\ kg^{-1}\ s^{-1}$	[2]
k_{BP}	bacterial inactivation via radicals	1.9×10^7	$m^3\ kg^{-1}\ s^{-1}$	[13]
I	intensity of UV light	2	W m^{-2}	[14,15]

Table 1: Default values for physical constants used for our model, determined from literature.

For the values of k_{BC} and k_{BI}, we primarily looked at the experimental database from Mofidi et al. [2]. We look at experimental results from 2020 onwards, with pH = 7–8, T = 20–24 °C, and bacterial reduction $\geq 3 \log_{10}$. For chlorine, we divide the $\log_{10}$ reduction by the time-concentration exposure `CT` (concentration × time). For UV, we take the dose directly from column `UV_D`. We take the 95% percentile of values, since we want to target most bacteria, but ignore ultra-resistant strains, as they can take up to 4 orders of magnitude more dose to inactivate, and we are trying to model a representative system (and assume that chlorine-resistant pathogens are still inactivated by UV light, and vice-versa).

The chlorine photolysis rate k_{IC} was calculated assuming direct irradiation by monochromatic ultraviolet light at 254 nm and pH 7.5, given a quantum yield for first-order radicals produced by hypochlorous acid HOCl (41% of free chlorine) and hypochlorite ions OCl^- (59% of free chlorine). These values were then used to calculate k_P and k_{BP} by looking at individual conversion rates of chlorine $Cl^{-}\bullet$ and hydroxyl $HO\bullet$ radicals, how they are produced from HOCl and OCl^-, and finally the ratios were converted to mass concentrations using molar mass.

Moreover, as we have noted in Section 3.3, we make assumptions about pH and temperature. In reality, many of the constants we use are dependant on temperature, most notably pathogen reproduction rates k_B. If the water treatment plant has significantly different or varying temperatures or pH, we would recommend looking at the literature to review all the k values, since they can vary significantly.

3.5 Implementation

```
from numpy           import array, exp
from scipy.integrate import solve_ivp
from parameters      import kC, kP, kBC, kIC, kBP, I, C0, T

def system(t: float, y: [float, float], δs: tuple(float)):
    b, p = y                                  # extract current b, p
    c = exp(-δs[3] * t)                       # calculate current c
    return array([                            # calculate derrivatives
        b * (δs[1] - δs[2] * p - c)         ,   # db/dt
        δs[4] * c - p * (δs[5] + δs[6] * b) ])  # dp/dt

def simulation(C0: float, B0: float, I: float, kB: float):
    δs = (0,                                  # δ values, 1-indexed
             (kB - kBI * I) / (kBC * C0)  ,  # δ_1
              kBP           / kBC         ,  # δ_2
             (kC + kIC * I) / (kBC * C0)  ,  # δ_3
              kIC * I       / (kBC * C0)  ,  # δ_4
              kP            / (kBC * C0)  ,  # δ_5
              kBP           / (kBC * C0)  )  # δ_6

    t_max = 32.0 * kBC * C0 * T    # simulate after the treatment time
    res = solve_ivp(method="Radau", atol=1e-14, rtol=1e-12, max_step=1.0,
                    fun=system, t_span=(0.0, t_max), args=(δs,),
                    y0=array([1.0, 0.0]))    # b(0) = 1, p(0) = 0

    ts = res.t                      # list of times
    cs = exp(-δs[3] * ts)           # list of chlorine concentrations
    bs, ps = res.y                  # lists of pathogen and radical conc.
    return ts, bs, cs, ps
```

Listing 1: Implementation of the dimensionless model and solver.

The values in Table 1 tell us that the system spans approximately 12 orders of magnitude. Therefore, we needed to use a method that was numerically stable even at very low values. This was also one of our motivations for rescaling the model, as it allowed us to not have to worry about hitting numerical precision.

Since our problem was stiff, we decided to use a Radau IIA solver—an implicit Runge-Kutta method of order 5 accuracy—with a maximum step size of $\tau k_{BC} C_0 = 1$ and absolute and relative tolerances of 10^{-14} and 10^{-12} respectively. We used both Matlab and Python for prototyping, with the final model being implemented using SciPy's `scipy.integrate.solve_ivp` (with Radau method) that follows Hairer and Wanner [16], as can be seen in Listing 1.

4 Results

The first thing we can look at is whether our system behaves as we would expect. For our experiments, we were always targeting pathogen reductions $\geq 3 \log_{10}$, i.e. reducing the population by at least 99.9%. In Figure 2, we can see that the time it takes for bacteria to reduce by $3 \log_{10}$ is in the order of 10–100 seconds (around a minute), which is consistent with what we would expect from literature ([1,2]). Furthermore, we see that photolysed radicals hover around 0, which is expected, since the equilibrium concentrations of radicals are about 7 orders of magnitude smaller than the concentration of chlorine, and here they share the scale C_0.

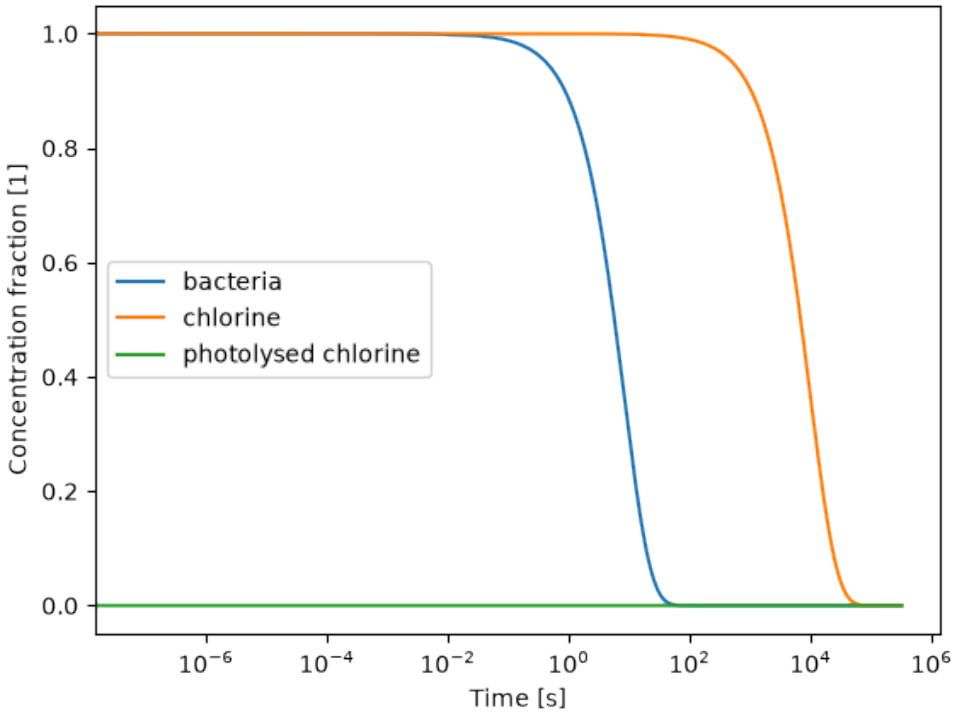


Figure 2: Run of the simulation with default values from Table 1.

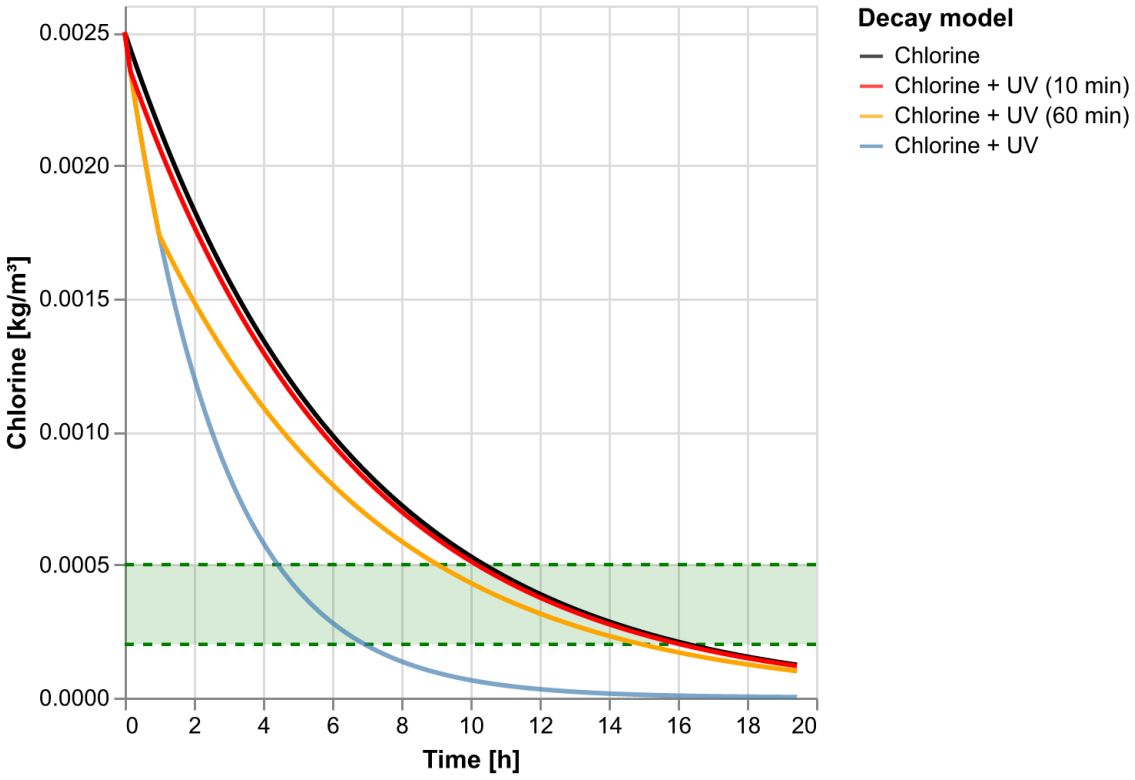


Figure 3: Concentrations of chlorine in water, comparing 4 decay models. Initial and safe residual amounts (drinking water; shown in green) are from Table 1.

Moreover, we can see that using World Health Organisation's recommended chlorine concentrations [3], we have a residual chlorine level for 10^5–10^6 seconds (i.e. on the order of a day). We can investigate this better by looking at chlorine directly. Shown in Figure 3, we can see that safe residual levels of chlorine remain in the system for several hours after treatment, which is what we would expect.

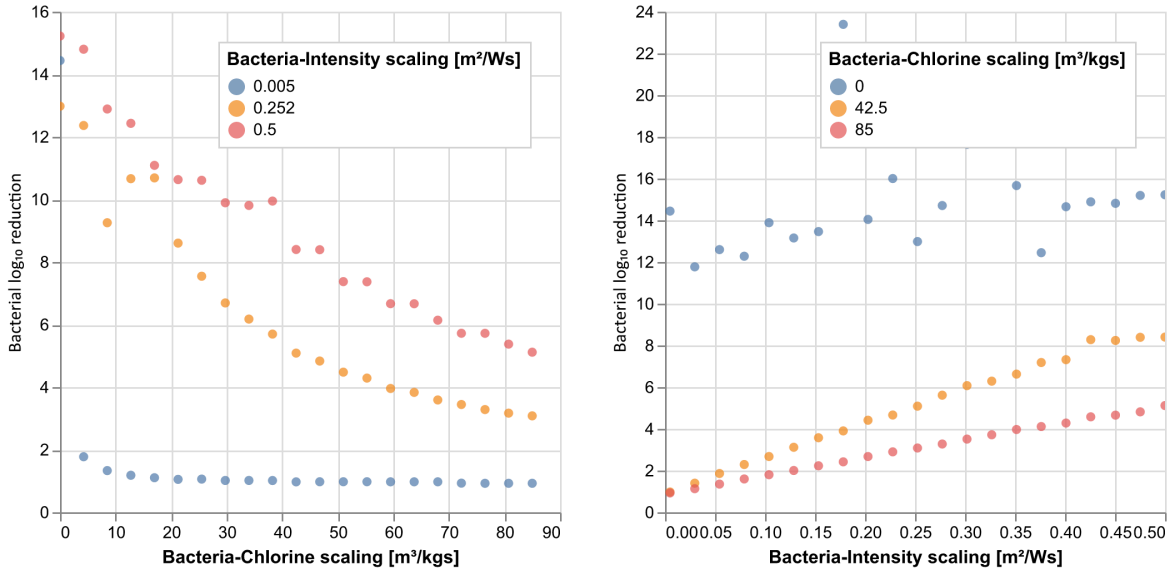


Figure 4: Pathogen reduction with varied k_{BC} and k_{BI}.

Next, we investigate how pathogen reduction correlates with the reactivity to chlorine and irradiation (Figure 4). Since our timescale (and therefore the runtime of the experiment) is set by k_{BC}, we expect larger values to result in shorter runtimes and therefore less reduction. As for pathogen-irradiation scaling, we expect a linear scaling.

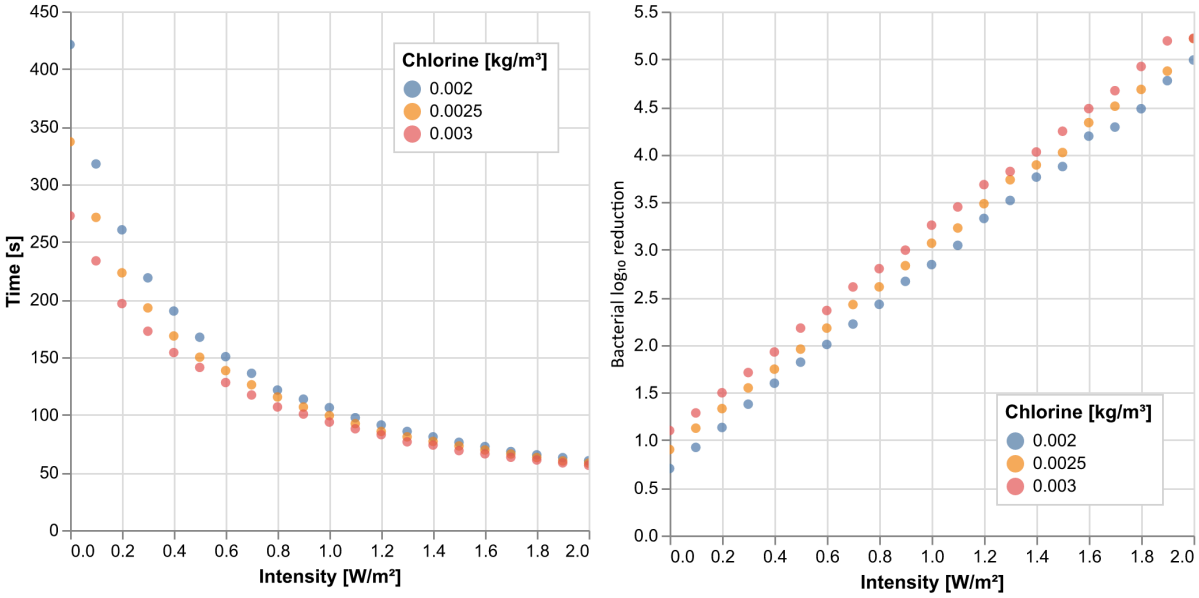


Figure 5: Time for $3\log_{10}$ pathogen reduction (LEFT) and pathogen reduction at $t = 100$ s (RIGHT), varying UV light intensity.

Last, we look at how the pathogen population behaves under varying amounts of UV light intensity. In Figure 5, we can see that a 90% reduction in intensity takes approximately 4 times as long to achieve a $3\log_{10}$ reduction in pathogens. Effectively, this sets an upper bound on how long unsanitised water needs to spend in a reactor, given that some (or indeed most) pathogens will not be subject to the full dose of UV fluence, as discussed in Section 3.3.

5 Conclusion

In summary, we implemented a model for combined ultraviolet light and chlorine treatment of filtered water, assuming homogeneity and isotropy. Taking experimental values from literature (Table 1), we simplify our model by neglecting chlorine decay. We simulate the system numerically using Radau methods for stability and 5th order accuracy, since our problem is stiff.

The final model is consistent with experimental results, and could be used to estimate water treatment systems, given a set of input parameters, provided that our assumptions hold. We also detail how the model can be adjusted in Section 5.1, and suggest experiments that would make the model more accurate in Section 5.2.

5.1 Limitations and further works

Model fluid dynamics

Our main assumption relies on the system being isotropic and homogeneous, as well as having fully dissolved chlorine at time $t = 0$. In practice, we know that this assumption does not hold based on studies that model the water treatment system ([17–19]), and that about 10% of the pathogens do not get exposed to sufficient disinfectant to be inactivated.

To study the model dynamics, this would require including a diffusion and advection term for all three spatial directions, as well as modelling the actual reactor or flow-through system. Due to the time limitations, and to avoid making further assumptions about the system via its geometry, we did not pursue this during the project.

However, knowing the distribution of chlorine and radiation dose received over the pathogen population would allow us to make meaningful estimates by adjusting the simulation constants.

Robust scaling functions

Another assumption that does not fully hold is temperature and pH being constant. Even within a single treatment plant, seasonal temperature variance can make the model predictions differ from reality. Since most of the values in Table 1 depend on temerature and pH, finding more robust values that can be calculated, given an input (T, pH) pair could make the predictions much more robust.

Furthermore, we assume that the water only contains H_2O molecules, but even perfectly disinfected water still includes minerals and other trace materials. These also react with chlorine, and are introduced to the system if chlorine is added to the system by adding hypochlorous salts. Therefore, modelling mineral contents of the water might be interesting.

Toxic byproducts

Although we have touched on this issue in Section 3.3, we believe that modelling the toxic byproducts created by chlorine reacting with residual organic matter is crucial to ensure safety. Therefore, we suggest extending the model to also account for both toxic and carcinogenic elements, which should not be too difficult as the dependance is only on chlorine and its radicals. However, literature on conversion rates is sparse, and we were not able to find sufficiently rigirous details.

Robust photolysed chlorine model

Our system only models first-order chlorine radicals. However, these compounds can react further, creating second- and higher-order radical compounds that it would be interesting to model.

Robust bacterial/viral/protozoal model

As detailed in Section 3.4, we are currently using aggregate values, thereby treating all pathogenic matter as a single monolith. However, data is usually available with per-pathogen granularity (at least for chlorine and UV, photolysis radical data is sparse). Therefore, it should be possible to separate the model into separate B_i per population type, ensuring that both common bacteria and resistant outliers can be modelled independently.

Downstream requirements

The World Health Organisation sets limits on the minimum and maximum residual chlorine to ensure that residual pathogens and other matter downstream can be neutralised. However, if we know the specific requirements (such as how long the water takes to get from treatment plant to the consumer), we would be able to model the dosage more precisely.

5.2 Helpful experimental data

Most of the experimental data that would be helpful is plant-specific—comprehensive data is available for how bacteria and viruses interact with chlorine and UV, but without knowing what pathogens are present in the water supply, we can only work with aggregates. Similarly, knowing the exact temperature and pH levels, when combined with suggested extensions from Section 5.1, can make the model very accurate.

The one set of data that is not readily available is the reaction between pathogens and chlorine radicals. Therefore, experiments that would give us more detailed data on this reaction would help us quantify how much more effective combining these two methods are numerically (as of today, we understand the mechanisms and have empirical data that combined treatment is better, but the specifics vary [6]). A possible experiment would be to test UV-resistant pathogens in a chlorine-only, and chlorine-UV treatment to be able to isolate the contribution from photolysed radicals.

Bibliography

[1] Kishimoto N 2019 State of the Art of UV/Chlorine Advanced Oxidation Processes: Their Mechanism, Byproducts Formation, Process Variation, and Applications *Journal of Water and Environment Technology* **17** 302–35

[2] Mofidi A, Linden K, Bedell E, Vetrovs A, Friedman M, Won D, Deem S, Rice N, Jahne M, Nicholas T and Anderson A 2025 Development of a disinfection efficiency database for bacterial inactivation: A systematic literature review for selected water treatment technologies *Critical Reviews in Environmental Science and Technology* **55** 1413–29

[3] World Health Organization 2022 *Guidelines for Drinking-water Quality: Fourth Edition Incorporating the First and Second Addenda* (Geneva: World Health Organization)

[4] Department for Environment Food & Rural Affairs 2026 Bathing Water Quality

[5] Strategy N W Q M 2026 Australian Drinking Water Guidelines

[6] Gámiz J, Grau A, Martínez H and Bolea Y 2020 Automated Chlorine Dosage in a Simulated Drinking Water Treatment Plant: A Real Case Study *Applied Sciences* **10**

[7] Pollard P 2003 Wastewater: Growth Kinetics of Microorganisms

[8] García-Ávila F, Sánchez-Alvarracín C, Cadme-Galabay M, Conchado-Martínez J, García-Mera G and Zhindón-Arévalo C 2020 Relationship between chlorine decay and temperature in the drinking water *MethodsX* **7**

[9] Zhang H, Zhu Z, Zhou X, Tang B, Yu J, Zhang B, Ma J and Yang T 2022 Degradation of ibuprofen by the UV/chlorine/TiO2 process: Kinetics, influencing factor simulation, energy consumption, toxicity and byproducts *Chemical Engineering Journal* **450** 137849

[10] Masjoudi M and Mohseni M 2022 Photochemical Properties of Free Chlorine and Inorganic Chloramines at 185 nm for VUV/UV/Chlorine Advanced Oxidation *ACS ES&T Water* **2** 769–77

[11] Feng Y, Smith D W and Bolton J R 2007 Photolysis of aqueous free chlorine species (HOCl and OCl^-) with 254 nm ultraviolet light *Journal of Environmental Engineering and Science* **6** 277–84

[12] Mutschlechner M, Chisté D, Hauptmann D and Schöbel H 2024 Ultraviolet (Spot)light on Water Treatment: Targeting Inactivation Efficiency and Stress Responses of Gram-Positive and Gram-Negative Bacteria Using UV-B and UV-C LEDs *Water* **16** 2028

[13] Wang W-L, Zhang X, Wu Q-Y, Du Y and Hu H-Y 2017 Degradation of natural organic matter by UV/chlorine oxidation: Molecular decomposition, formation of oxidation byproducts and cytotoxicity *Water Research* **124** 251–8

[14] Zyara A M, Torvinen E, Veijalainen A-M and Heinonen-Tanski H 2016 The Effect of UV and Combined Chlorine/UV Treatment on Coliphages in Drinking Water Disinfection *Water* **8**

[15] LeChevallier M W 2004 Water treatment and pathogen control

[16] Hairer E and Wanner G 1996 *Solving Ordinary Differential Equations II. Stiff and Differential-Algebraic Problems* vol 14

[17] Bolton J R and Linden K G 2003 Standardization of Methods for Fluence (UV Dose) Determination in Bench-Scale UV Experiments *Journal of Environmental Engineering* **129** 209–15

[18] Chiu K, Lyn D A, Savoye P and III E R B 1999 Integrated UV Disinfection Model Based on Particle Tracking *Journal of Environmental Engineering* **125** 7–16

[19] Qiang Z, Li M, Bolton J R, Qu J and Wang C 2013 Estimating the fluence delivery in UV disinfection reactors using a 'detector-model' combination method *Chemical Engineering Journal* **233** 39–46